

Math 075 Homework 5

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Exercise 2.5.1

Problem

For each of the following elementary matrices, describe the corresponding elementary row operations.

Part (a)

$$E = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution (a)

$$E^{-1} = \begin{bmatrix} 1 & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Exercise 2.5.2

Problem

In each case find an elementary matrix E such that $E = BA$.

Part (a)

$$A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$$

Solution (a)

$$A \longrightarrow B$$

$$B = EA \longrightarrow \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$$
$$E = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

Exercise 2.5.3

Problem

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$.

Part (a)

Find elementary matrices E_1 and E_2 such that $C = E_2 E_1 A$

Solution (a)

$$\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_1} \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$C = E_2 E_1 A \longrightarrow C = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$$

Exercise 2.5.8

Problem

In each case factor A as a product of elementary matrices.

Part (a)

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

Solution (a)

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \longrightarrow E_1(A) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$E_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \longrightarrow E_2(E_1 A) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$E_3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \longrightarrow E_3(E_2 E_1 A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$E_3 E_2 E_1 A = I$$

$$A = E_1^{-1} E_2^{-1} E_3^{-1} I$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Exercise 2.6.1

Problem

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation.

Part (a)

Find $T \begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix}$ if:

$$T \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$T \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

Solution (a)

Let:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix} = a\mathbf{v}_1 + b\mathbf{v}_2$$

$$\begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$\begin{cases} a + 2b = 8 \\ 0a + b = 3 \\ a + 3b = 7 \end{cases}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = RREF \left(\begin{bmatrix} 1 & 2 & 8 \\ 0 & 1 & 3 \\ -1 & 3 & 7 \end{bmatrix} \right) \longrightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$T \begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix} = T(2\mathbf{v}_1 + 3\mathbf{v}_2) = T(2\mathbf{v}_1) + T(3\mathbf{v}_2)$$

$$T \begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix} + \begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

$$T \begin{bmatrix} 8 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

Exercise 2.6.2

Problem

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation.

Part (a)

Find $T \begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix}$ if:

$$T \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \quad T \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

Solution (a)

Let:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = a\mathbf{v}_1 + b\mathbf{v}_2$$

$$\begin{cases} a + 0b = 1 \\ a - b = 3 \\ 0a + b = -2 \\ -a + b = -3 \end{cases}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = RREF \left(\begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 3 \\ 0 & 1 & -2 \\ -1 & 1 & -3 \end{bmatrix} \right) \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = T(\mathbf{v}_1 - 2\mathbf{v}_2) = T(\mathbf{v}_1) + 2T(\mathbf{v}_2)$$

$$T \begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \\ 2 \end{bmatrix}$$

$$T \begin{bmatrix} 1 \\ 3 \\ -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 12 \\ 3 \\ 1 \end{bmatrix}$$

Exercise 2.6.3

Problem

In each case assume that the transformation T is linear, and use Theorem 2.6.2 to obtain the matrix A of T .

Part (a)

$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the reflection in the line $y = -x$

Solution (a)

$$\begin{aligned} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \\ A &= \begin{bmatrix} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} & T \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} \longrightarrow A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \end{aligned}$$

Exercise 2.6.7**Problem**

In each case show that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is not a linear transformation.

Part (a)

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} xy \\ 0 \end{bmatrix}$$

Solution (a)

Let:

$$\mathbf{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

T is a linear transformation if:

$$\begin{aligned} T(\mathbf{u} + \mathbf{v}) &= T(\mathbf{u}) + T(\mathbf{v}) \\ T \left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right) &= T \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + T \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \\ T \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} &= T \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + T \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}\begin{bmatrix} (x_1 + x_2)(y_1 + y_2) \\ 0 \end{bmatrix} &= \begin{bmatrix} x_1 y_1 \\ 0 \end{bmatrix} + \begin{bmatrix} x_2 y_2 \\ 0 \end{bmatrix} \\ \begin{bmatrix} (x_1 + x_2)(y_2 + y_2) \\ 0 \end{bmatrix} &\neq \begin{bmatrix} x_1 y_1 + x_2 y_2 \\ 0 \end{bmatrix}\end{aligned}$$

T does not satisfy the property of additivity. Therefore, it is not a linear transformation.